How AdamW and Muon Train a Small Language Model

Next steps (ii) and (iii) of the checkpoint

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Summary. The Step 2 runs reproduce on a new cluster. (ii) The update cosine comes from the rule: it differs on pure noise. Stable rank and $\sigma_1/\sigma_{\text{median}}$ are ordered by the rule on real gradients; the spectral norm is not. (iii) Weight decay does not explain Muon's speed, and without orthogonalization Muon ends worse than AdamW. Whether the spectra cause the speed is open.

1 Setup and reproduction

TinyStories-10M, the checkpoint's settings: AdamW at $2 \cdot 10^{-3}$, Muon at 0.01, 10,000 steps, data orders 0 to 2. Run on FAS RC Cannon. Spectral metrics are means over the 36 block weight matrices, as in the checkpoint. The rerun agrees with the checkpoint (Table 1).

Table 1: Rerun (checkpoint in brackets); the last two rows at training loss 1.81.

	AdamW	Muon
final validation loss	1.339 to 1.345 (1.344)	1.308 to 1.310 (1.309)
step reaching AdamW's final validation loss		7250 to 7750 (7500)
stable rank of W	32.7 ± 0.2 (33.0)	86.3 ± 0.5 (86.2)
active MLP units	0.120 ± 0.001 (0.120)	0.222 ± 0.001 (0.222)

2 (ii) Direct result of Muon's updates, or emergent in training?

2.1 Rules without learning

Each rule updates one block matrix of each shape for 10,000 steps at the runs' learning rates, on synthetic gradients: noise, or a rank-4 direction plus noise of equal norm. Nothing is learned (Fig. 1).

Update cosine follows the rule. On noise, consecutive updates have cosine 0.90 under AdamW and 0.66 to 0.76 under Muon, close to the real runs (0.89 and 0.74 at loss 1.81).

The spectra do not. On noise, neither rule moves the stable rank of W. With the rank-4 signal both fall to about its rank, AdamW's within a few hundred steps and Muon's by the end (step 1000: 3.7 against 9.9).

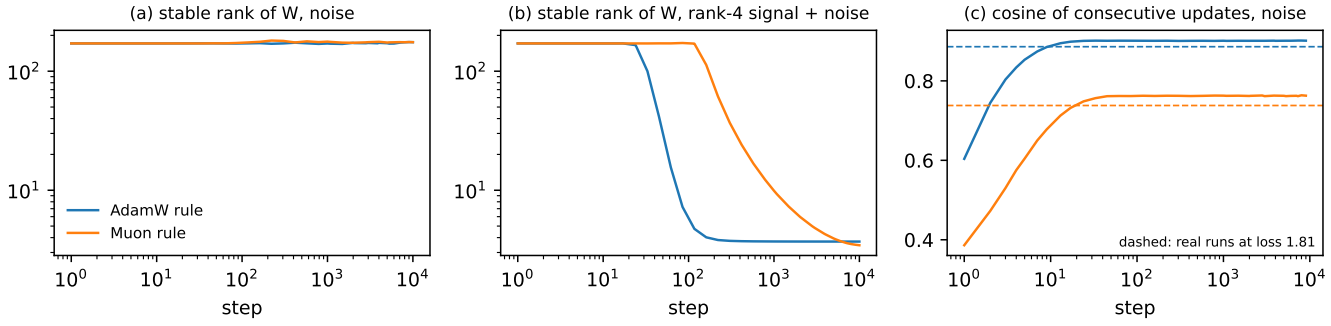


Figure 1: Each rule on synthetic gradients, MLP input matrix.

2.2 Ghosts on real gradients

Each run carries a *ghost*: a copy of the initial weights updated by the other optimizer's rule on the run's own gradients. The ghost never feeds back into the run. With the two runs this makes a 2×2 design, rule $\times$ gradient stream (Table 2, Fig. 2).

The ordering follows the rule. Muon's rule on AdamW's gradients ends at stable rank 87, against 40 for the AdamW run and 75 for the Muon run. AdamW's rule on Muon's gradients ends at 13. $\sigma_1/\sigma_{\text{median}}$ and update cosine go the same way. Both ghosts go further than the run with their rule (87 against 75, 13 against 40): the gradient stream matters too.

The spectral norm does not. At step 10,000, on AdamW's gradients Muon's rule gives 10.9 and AdamW's 5.9; on Muon's gradients, 7.5 and 7.8. Which rule is larger depends on the gradient stream.

Not shown. A ghost is not evaluated as a model, so this says nothing about $\lambda_{\max}$, MLP sparsity or the loss.

Table 2: The 2×2 design at step 10,000.

rule	gradient stream	stable rank	$\sigma_1/\sigma_{\text{median}}$	update cosine
AdamW	AdamW (run)	39.9 to 41.4	5.80 to 6.19	0.89 to 0.90
AdamW	Muon (ghost)	12.6 to 13.0	15.6 to 16.0	0.89
Muon	Muon (run)	75.3 to 75.5	2.89 to 2.90	0.71
Muon	AdamW (ghost)	87.0 to 87.2	2.77 to 2.80	0.70

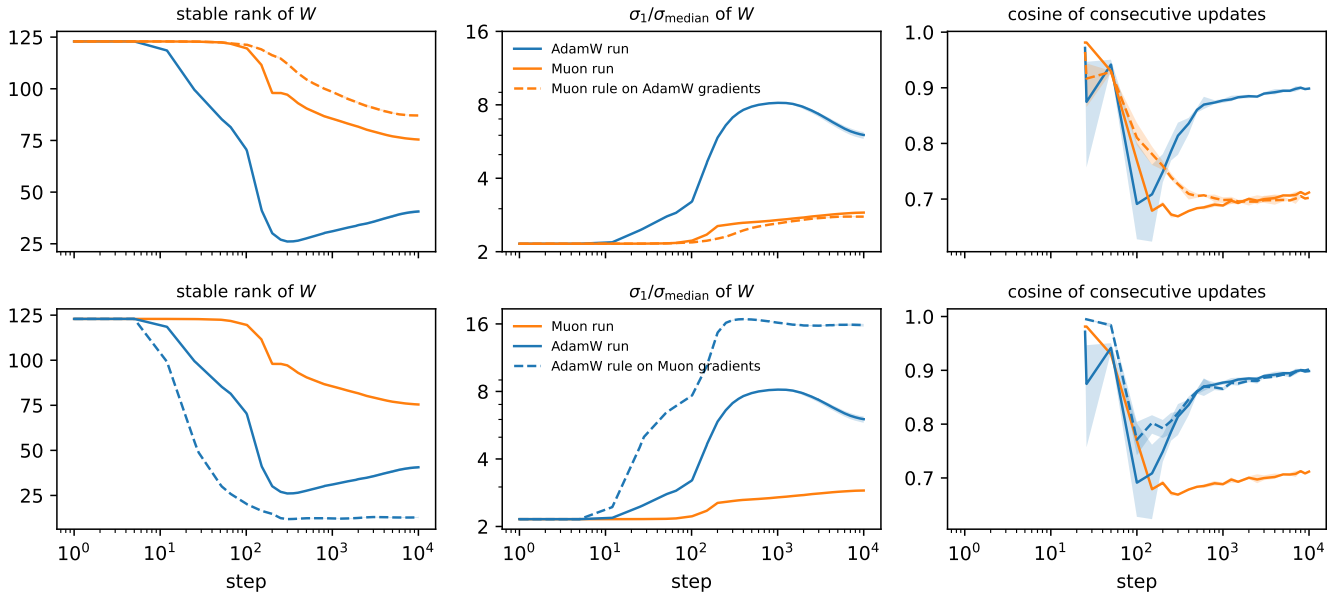


Figure 2: Ghosts (dashed) against runs.

3 (iii) Do the differences explain Muon's faster training and different path?

3.1 What the speed needs

Three ablations: AdamW without weight decay, Muon with weight decay 0.02, and Muon without orthogonalization, its momentum rescaled to an orthogonal update's norm (Table 3).

Weight decay is not the reason. Muon has none and is already faster. Removing it from AdamW costs 0.005 to 0.012 in final loss; adding it to Muon gains 0.006 to 0.011, and Muon still reaches AdamW's final loss at step 7750 (data order 0).

Without orthogonalization Muon ends worse than AdamW. This holds at every learning rate tried (last three rows).

Table 3: Final validation loss at step 10,000.

run	data order		
	0	1	2
AdamW (baseline)	1.339	1.341	1.345
AdamW, weight decay 0	1.346	1.353	1.350
Muon (baseline)	1.308	1.309	1.310
Muon, weight decay 0.02	1.297	1.298	1.304
Muon, no orthogonalization, lr 0.01	3.729	2.943	1.389
Muon, no orthogonalization, lr 0.003	1.454		
Muon, no orthogonalization, lr 0.03	4.078		

3.2 Dose-response

Muon's update with its singular values raised to a power p: $p = 0$ is exact orthogonalization (an exact SVD, not Muon's Newton-Schulz approximation), $p = 1$ is none. Muon at learning rate 0.01, data order 0 (Table 4).

Speed and spectra move together with p. As p grows, the run reaches loss 1.81 later, and its stable rank and $\sigma_1/\sigma_{\text{median}}$ move toward AdamW's.

$p = 0.75$ **on other data orders.** On orders 0 and 1 the stable rank matches AdamW's (33.4 to 33.5 against 32.4 to 32.9) and loss 1.81 comes at about the same step (1379 to 1394 against 1426 to 1469); final validation loss is worse (1.354 to 1.357 against 1.339 to 1.345). Order 2 stalls from about step 60 and ends at 1.659. At learning rate 0.003 (order 0), $p = 0.75$ has stable rank 25.6, below AdamW's, and reaches loss 1.81 at step 1298 against AdamW's 1469; $p = 1$ reaches it at step 2926.

Not shown. p changes both, so the table cannot say which causes which.

Table 4: Muon with singular values raised to p, data order 0; step and spectra at training loss 1.81.

run	step at loss 1.81	stable rank	$\sigma_1/\sigma_{\text{median}}$	final val. loss
Muon (Newton-Schulz)	846	86.6	2.68	1.308
$p = 0$ (exact SVD)	827	88.2	2.64	1.306
$p = 0.25$	949	63.8	3.39	1.321
$p = 0.5$	1230	41.3	5.14	1.341
$p = 0.75$	1394	33.5	7.61	1.357
AdamW	1469	32.9	7.99	1.339